

Supporting Information

Coordinated cation transport in $\text{Ti}_3\text{C}_2\text{T}_x$ MXene membranes

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S1. Detailed Experimental Methods

Synthesis of Ti_3AlC_2 MAX Precursor

The MAX phase precursor was synthesized in house by ball milling TiC (Alfa Aesar, 99.5%, 2 μm), Ti (Alfa Aesar, 99.5%, 325 mesh), and Al (Acros Organics, 99%, 200 mesh) powders in a 2:1.25:2.2 molar ratio. This ratio incorporates excess Al and has been reported to improve MXene oxidation stability.¹ The mixture was milled intermittently at 300 rpm for 7 hours (12 cycles, each 5 minutes followed by 30 minutes pause) in steel-lined zirconia milling jars (Fritsch Pulverisette). Each jar contained 12 g of precursor powders, 12.5 mL isopropyl alcohol, and approximately 100 g milling media (yttria-stabilized zirconia, 3 mm diameter). The milled precursor powder was then dried in a vacuum oven at 50 °C and sieved through a 140-mesh (106 μm) particle sieve.

The milled precursor powder was packed into an alumina crucible and placed into a tube furnace. The precursor powders were heated to 1380 °C at a rate of 3 °C/min, then held at 1380 ° for 2 h, under constant argon gas flow. After cooling, the sintered block of Ti_3AlC_2 was mechanically scraped to remove TiO_2 from the surface, ground into fine powder using a mortar and pestle, and sieved through a 325-mesh (45 μm) particle sieve.

$\text{Ti}_3\text{C}_2\text{T}_x$ MXene Synthesis

Aqueous dispersions of single-layer $\text{Ti}_3\text{C}_2\text{T}_x$ MXene were synthesized using the MILD method.² For each synthesis batch, 1.6 g of dry LiF powder (Alfa Aesar, 97%) was added to 20 mL of 9 M HCl and stirred for at least 5 minutes at 300 rpm to generate HF etchant in situ. Then, 1.0 g of the sieved MAX precursor was gradually added, and the solution was stirred for 24 h at 35 °C and 300 rpm.

After 24 h, the etched MXene was collected into a centrifuge tube and centrifuged for 7 minutes at 3500 rpm. The supernatant was discarded, and the MXene sediment was washed with deionized water. The centrifugation and washing steps were repeated until the

supernatant reached pH 6 or higher. The neutral MXene dispersion was then bath sonicated in ice for 20 minutes to increase flake delamination and improve yield.

After sonication, the MXene dispersion was centrifuged for 45 minutes at 3500 rpm, and the light-colored supernatant (very small MXene flakes) was discarded. The remaining MXene sediment was thoroughly re-dispersed in deionized water and centrifuged again for 10 minutes. This time, the dark-colored supernatant (typical single-layer MXene flakes) was collected. Re-dispersion and single-layer MXene collection were repeated until the supernatant became dilute. The collected single-layer MXene dispersion was stored in a large bottle and refrigerated.

Preparation of PVDF-Supported $\text{Ti}_3\text{C}_2\text{T}_x$ Membranes

First, to test the $\text{Ti}_3\text{C}_2\text{T}_x$ dispersion's concentration, ~ 10 mL of the dispersion was filtered via vacuum-assisted filtration (VAF) onto a porous PVDF membrane (EMD Millipore Durapore, hydrophilic, $0.45\text{ }\mu\text{m}$ pore size, 47 mm diameter) and dried under vacuum for at least 12 h. The mass change was recorded and used to determine the initial concentration of MXene (typically 1-2 mg/mL). The dispersion was then diluted to 0.4 mg/mL by adding deionized water.

Supported MXene membranes were fabricated via VAF of the diluted MXene dispersion onto the PVDF membranes. 25 mL of 0.4 mg/mL dispersion was used per membrane; the filtered membranes were dried under vacuum for at least 12 h and kept on top of the PVDF supports. This procedure produced MXene membranes with a consistent MXene mass of ~ 10 mg and a MXene active layer thickness of $\sim 3\text{--}4$ mg. Thickness was measured using a micrometer; the thickness of a bare PVDF membrane was subtracted from the total membrane thickness. The dry membranes were tested for pinholes using a flashlight. If pinholes were found, the membrane was discarded and not used for permeation testing.

Membrane Permeation Testing

The dry MXene membranes were inserted into an H-cell setup for diffusion testing (Fig. S1). Membranes were sandwiched between two rubber gaskets (inner diameter 15 mm, outer diameter ~ 26 mm) and gently cut to size. The gasket-sealed membranes were clamped between two sides of a glass H-cell and wrapped with paraffin film. 50 mL deionized water was poured into the side of the H-cell facing the PVDF support. 50 mL salt solution was poured into the side facing the MXene active layer. Both sides of the H-cell were stirred at 100 rpm to minimize the effects of concentration polarization. This configuration was kept constant for all tests.

For single-salt permeation tests, 0.2 M solutions of LiCl, NaCl, and CaCl₂ were used. For multi-salt mixture permeation tests, each cation's concentration was still kept at 0.2 M, resulting in total concentrations of 0.4 M. 0.2 M LiCl + 0.2 M NaCl, 0.2 M LiCl + 0.2 M CaCl₂, and 0.2 M NaCl + 0.2 M CaCl₂ were used.

The concentration of ions in the permeate side of the H-cell was monitored every hour for 5 hours to calculate ion flux and permeability. For single-salt solutions, a conductivity probe (Orion Duraprobe, Thermo Scientific) was used to detect the permeate conductivity, and readings were converted into concentration values. For multi-salt solutions, ion chromatography (940 Vario, Metrohm) was used to directly measure each ion's concentration. In both cases, calibration curves were made using stock solutions of each salt, and the measured conductivity values for each permeate solution were compared against the calibration curves. The change in ion concentration over time was normalized by membrane area, thickness, solution volume, and salt concentration to obtain each ion's permeability.

MXene Membrane Characterization

X-ray diffraction data were collected using a Bruker D8 diffractometer (Princeton Imaging and Analysis Center). After permeation testing, the membranes permeated with each salt solution were removed from the H-cell. First, each membrane was immediately measured in the

diffractometer while still covered in water (wet XRD). After this measurement, membranes were dried under vacuum for at least 12 hours and measured the next day (dry XRD).

Images of the MXene membranes were collected using scanning electron microscopy (Verios SEM, Princeton Imaging and Analysis Center). Cross-sectional images were collected by using a focused ion beam to mill a section of the membrane, then imaging the cut section (Helios FIB-SEM, Princeton Imaging and Analysis Center).

Molecular Dynamics Simulations

Molecular dynamics (MD) simulations were performed using LAMMPS³ to investigate cation transport and selectivity (Li^+/Na^+ , $\text{Li}^+/\text{Ca}^{2+}$, $\text{Na}^+/\text{Ca}^{2+}$) through stacked $\text{Ti}_3\text{C}_2\text{OOH}$ MXene membranes. The simulation system consisted of a four-layer MXene membrane (inter-layer spacing: 6.7–7.3 Å for wet state and 3.2–5.8 Å for dry state, determined by XRD in Figure 3) positioned between an ion-containing feed region (0.2 M single-salt solutions or 0.4 M binary mixtures with equimolar cation concentrations) and a pure water permeate region (Figure S3). Pressure gradients were applied through rigid graphene pistons at both ends of the $377 \times 54.1 \times 42.55\text{--}46.25$ Å³ simulation box (periodic boundary conditions in all directions), with optimized pressures of 450 MPa (single-salt) and 900 MPa (binary mixtures) for wet membranes, and 550 MPa (single-salt) and 1100 MPa (binary mixtures) for dry membranes. To eliminate artificial interactions, a vacuum spacing of ~ 150 Å was maintained between pistons along the x-direction. Water density was constrained to 1.0 ± 0.1 g/cm³ with randomized initial ion distributions in the feed region.

The MXene membrane’s parameters were adopted from Zhu and Xiong,⁴ while the graphene piston parameters were sourced from Aluru et al.⁵ Ion interactions utilized parameters by Merz⁶ with the TIP3P-FB water model, employing Lorentz-Berthelot combining rules. Van der Waals interactions employed a 9.0 Å cutoff, and long-range Coulomb interactions were calculated using the PPPM method.⁷ The MXene layers remained constrained during simulations, while water molecules and ions were allowed full mobility.

Following energy minimization via the conjugate gradient algorithm, the system underwent sequential equilibration: 0.25 ns in the NVT ensemble (300 K, Nose–Hoover thermostat) with fixed pistons and membrane, followed by 0.25 ns equilibration after releasing piston constraints. Production runs (2.5–40 ns duration depending on membrane hydration state and ion species) were conducted under non-equilibrium NVT conditions with maintained pressure gradients using a 2 fs timestep.

Density Functional Theory Calculations

The structures of MXene were constructed by incorporating hydrated metal cations in the interlayer space of the crystal structure (Ti_3SiC_2 , space group $\text{P6}_3/\text{mmc}$).⁸ The termination group of -OOH was employed in this study as our previous work⁴ showed that MXene with this group is the most reasonable. A $4\times 4\times 1$ supercell and AB stacking was considered and hydrated metal cations of each Li, Na, and Ca were added with 70 water molecules into the MXene interlayer space. The resulting structure has the composition of $\text{M}_{0.063}\text{Ti}_3\text{C}_2\text{OOH}\cdot 2.19\text{H}_2\text{O}$ (where M is Li, Na, and Ca). Density functional theory (DFT) calculations were then performed using Vienna ab initio simulation package (VASP).^{9,10} Projector augmented wave potentials were used with a GGA functional for the exchange-correlation functional¹¹ along with optB86b-vdW of nonlocal vdW-DF for the dispersion corrections.¹² A k-point spacing of $0.04 \text{ \AA}^{-1}$ and plane wave energy cut off of 520 eV were applied. The convergence criteria of electronic energy and force in ionic relaxation were set to the values of 10^{-5} eV and 10^{-2} eV/ $\text{\AA}$, respectively. Maximum bond lengths were 2.35, 2.62, and 2.81 $\text{\AA}$ for Li-O, Na-O, and Ca-O for hydration numbers, which are 1.2 times the values reported in Pyykkö et al.¹³ The optimized structures from DFT calculations were visualized using VESTA software.¹⁴ Binding energies (E_{BE}) were calculated using Equation S1:

$$E_{BE} = E(\text{M}_{0.063}\text{Ti}_3\text{C}_2\text{OOH}\cdot 2.19\text{H}_2\text{O}) - E(8(\text{M}_{0.125}\cdot 5\text{H}_2\text{O})) - E(16(\text{Ti}_3\text{C}_2\text{OOH})) + E(5(\text{H}_2\text{O})) \quad (\text{S1})$$

where $E(\text{M}_{0.063}\text{Ti}_3\text{C}_2\text{OOH}\cdot 2.19\text{H}_2\text{O})$, $E(8(\text{M}_{0.125}\cdot 5\text{H}_2\text{O}))$, $E(16(\text{Ti}_3\text{C}_2\text{OOH}))$, and $E(5(\text{H}_2\text{O}))$ denote the total electronic energy of a hydrated metal cation within the MXene interlayer, a hydrated metal cation with 40 water molecules considering the second hydration shell, layered MXene with an empty interlayer space, and 5 water molecules, respectively. This binding energy represents the energy required to intercalate a hydrated metal cation from the bulk aqueous solution to the empty interlayer space of MXene framework. Note that it is defined as the average energy for one of the two interlayer spaces in each structure.

S2. Calculation of Ion Permeability and Selectivity

Ion diffusion experiments were conducted using diffusion H-cells (schematic shown below in Figure S1). For all experiments, conductivity was measured on the permeate side of the cell (initially DI water) every hour. Conductivity readings were obtained via (a) a conductivity probe, for single-salt solutions, and (b) ion chromatography, for binary mixtures.

For single-salt solutions, the conductivity probe reported the conductivity of the entire solution (including the cation and Cl^-). Conductivity readings for each salt were compared against a calibration curve made using known concentrations of that salt. This yielded the ion concentration in the permeate at each hour.

For binary mixtures, ion chromatography yielded conductivity readings for each individual cation. These readings were compared against additional calibration curves made from standard solutions of each salt.

In both cases, ion concentration at the end of five hours was converted to an average ion permeation rate (ion flux) using Equation S2:

$$J_i = \frac{C_i V}{A \Delta t} \quad (\text{S2})$$

where J_i represents the molar flux of species i (in mol/m²h), C_i is the concentration of species i (in mol/L), V is the cell volume (0.05 L), A is the membrane area in the cell (in m²), and Δt is the elapsed time (5 h). Data were inspected to ensure that the ion permeation trend was approximately linear with time (i.e., no sudden jump in concentration due to membrane failure).

Ion flux was converted to permeability using Equation S3:

$$P_i = \frac{J_i L}{C_{i,feed}} \quad (\text{S3})$$

where P_i represents the permeability of species i (in m²/h), J_i is the flux of species i (in mol/m²h), L is the thickness of the membrane (in m), and $C_{i,feed}$ is the concentration of species i in the feed (0.2 M = 200 mol/m³).

Finally, after determining each ion's permeability, the selectivity of each pair of cations was calculated by taking the ratio of both permeabilities (Equation S4):

$$S_{i/j} = \frac{P_i}{P_j} \quad (\text{S4})$$

S3. Error Propagation

At least 4 replicates were taken for each ion permeability measurement. Error bars for ion permeability represent the standard deviation of all replicate measurements. Since selectivity is the ratio of two ions' permeabilities, selectivity error bars use the formula for error propagation of a quotient (Equation S5):

$$\sigma_S = S * \sqrt{\frac{\sigma_A^2}{A^2} + \frac{\sigma_B^2}{B^2}} \quad (\text{S5})$$

where σ_S and S are the standard deviation and mean of the selectivity, σ_A and A are the standard deviation and mean of ion A's permeability, and σ_B and B are the standard deviation and mean of ion B's permeability.

Table S1: Comparison of XRD (002) component peak properties for membrane samples permeated with each ion mixture. Relevant variables include peak position, integrated intensity (area), and Gaussian and Lorentzian FWHM. The approximate fraction of each ion occupying the MXene interlayer was calculated by taking the ratio of component peaks' integrated intensities.

Sample	Component Peak	Position (°)	Integrated Intensity (° × a.u.)	Gaussian FWHM (°)	Lorentzian FWHM (°)	% Li	% Na	% Ca
Water	Li	6.44	1976	0.07	0.94	100	0	0
Li only	Li	6.52	2154	0.17	1.36	100	0	0
Na only	Li	6.50	323	0.12	1.23	28.1	71.9	0.0
Na only	Na	6.96	828	0.09	0.78			
Ca only	Li	6.46	1080	0.13	1.50	46.8	0.0	53.2
Ca only	Ca	5.78	1227	0.05	0.76			
Li+Na	Li	6.46	321	0.19	1.65	20.6	79.4	0.0
Li+Na	Na	7.00	1237	0.12	0.74			
Li+Ca	Li	6.44	391	0.18	1.50	34.9	0.0	65.1
Li+Ca	Ca	5.84	731	0.13	0.83			
Na+Ca	Li	6.50	289	0.16	0.91	22.5	31.5	46.0
Na+Ca	Na	6.96	406	0.18	1.46			
Na+Ca	Ca	5.80	593	0.10	0.84			

Table S2: Approximate elemental content of the MXene membrane permeated with the NaCl/CaCl₂ mixture. The membrane contains approximately 5 times as much Ca as Na by mass. Data were obtained from two replicate samples. EDS analysis was performed at the Center for Nanoscale Materials at Argonne National Laboratory.

Element	Mass %
Ti	58.7 ± 0.1
O	19.0 ± 3.0
C	10.0 ± 1.5
F	6.0 ± 0.7
Cl	3.8 ± 0.7
Ca	2.0 ± 0.1
Na	0.4 ± 0.1
Al	0.2 ± 0.0

Table S3: Interlayer spacing, hydration number in the first hydration shell (CN), and hydration radius ($r(\text{M-O}_w)$) in the first hydration shell in the optimized structures of $\text{M}_{0.06}\text{Ti}_3\text{C}_2\text{OOH} \cdot 2.19\text{H}_2\text{O}$ ($\text{M} = \text{Li}, \text{Na}, \text{and Ca}$) by DFT calculations. Interlayer spacing and CN are averaged over two interlayer spaces in each structure.

	Interlayer spacing (Å)	CN	$r(\text{M-O}_w)$ (Å)
Li	6.77	4	1.98
Na	6.92	5	2.41
Ca	6.95	6.5	2.40

Table S4: Coordination number (CN) of Li^+ , Na^+ , and Ca^{2+} in bulk solution and within the MXene membrane for both single-salt and binary-mixture systems under wet conditions. The values in parentheses represent the corresponding interlayer spacing, as determined by XRD in Figure 3a.

Single salt	Li (7.36 Å)		Ca (6.81 Å)		Na (6.58 Å)	
	CN in bulk	CN in membrane	CN in bulk	CN in membrane	CN in bulk	CN in membrane
	3.89	2.70	6.72	6.23	5.34	3.91
Binary mixture	Li/Ca (6.58 Å)		Na/Ca (6.81 Å)		Li/Na (6.75 Å)	
	Li	Ca	Na	Ca	Li	Na
CN in bulk	4.01	7.09	5.14	6.59	3.85	4.84
CN in membrane	3.49	6.27	4.63	6.37	3.12	4.62

Table S5: Coordination number (CN) of Li^+ , Na^+ , and Ca^{2+} in bulk solution and within the MXene membrane for both single-salt and binary-mixture systems under dry conditions. The values in parentheses represent the corresponding interlayer spacing, as determined by XRD in Figure 3b.

Single salt	Li (4.13 Å)		Ca (5.81 Å)		Na (3.21 Å)	
	CN in bulk	CN in membrane	CN in bulk	CN in membrane	CN in bulk	CN in membrane
	3.79	2.54	6.71	5.99	5.19	3.00
Binary mixture	Li/Ca (5.81 Å)		Na/Ca (5.81 Å)		Li/Na (3.21 Å)	
	Li	Ca	Na	Ca	Li	Na
	CN in bulk	CN in membrane	CN in bulk	CN in membrane	CN in bulk	CN in membrane
	3.81	6.88	5.44	6.97	3.76	5.42
	3.37	6.35	4.67	6.41	2.55	3.36

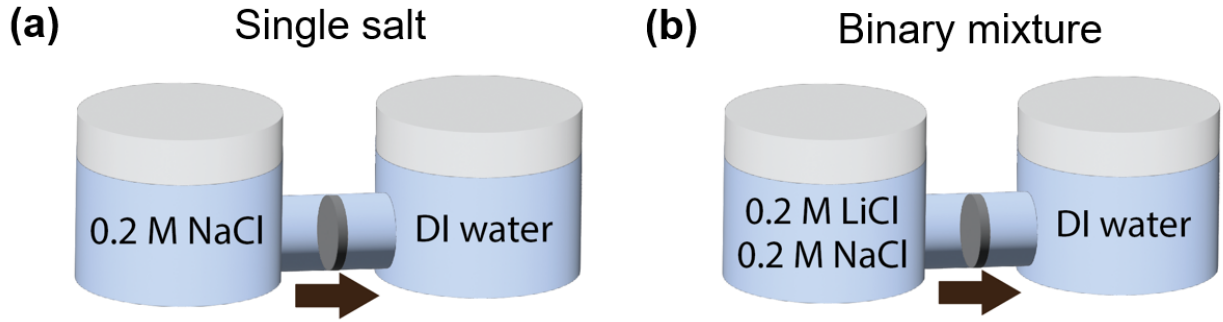


Figure S1: Schematic of the membrane permeation cell setup for (a) single-salt and (b) binary mixture testing.

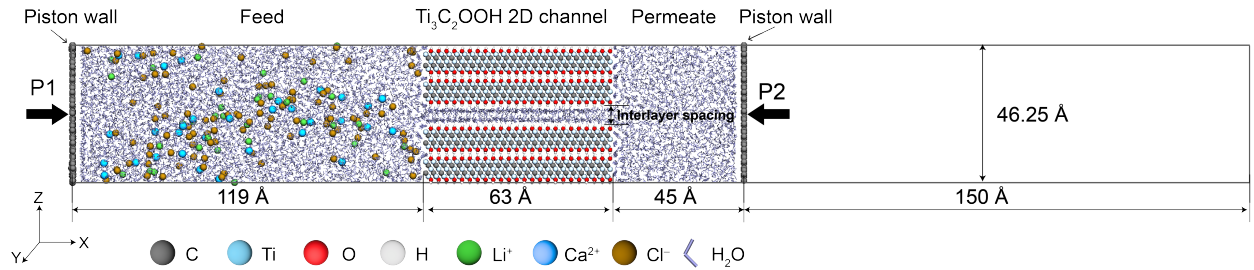


Figure S2: MD Simulation of ion transport through an MXene membrane with interlayer spacings of 6.7–7.3 Å in the wet state and 3.2–5.8 Å in the dry state, as determined by XRD in Figure 3. The $\text{Li}^+/\text{Ca}^{2+}$ binary mixture, with an interlayer spacing of 6.58 Å, is shown as a representative example. Two additional MXene sheets act as support layers. P1 and P2 denote the pressures applied to the piston walls at the feed and permeate sides along the x-direction. Colors: carbon (gray), titanium (light blue), oxygen (red), hydrogen (white), water (ice blue), ions (as in the legend), and graphene pistons (gray).

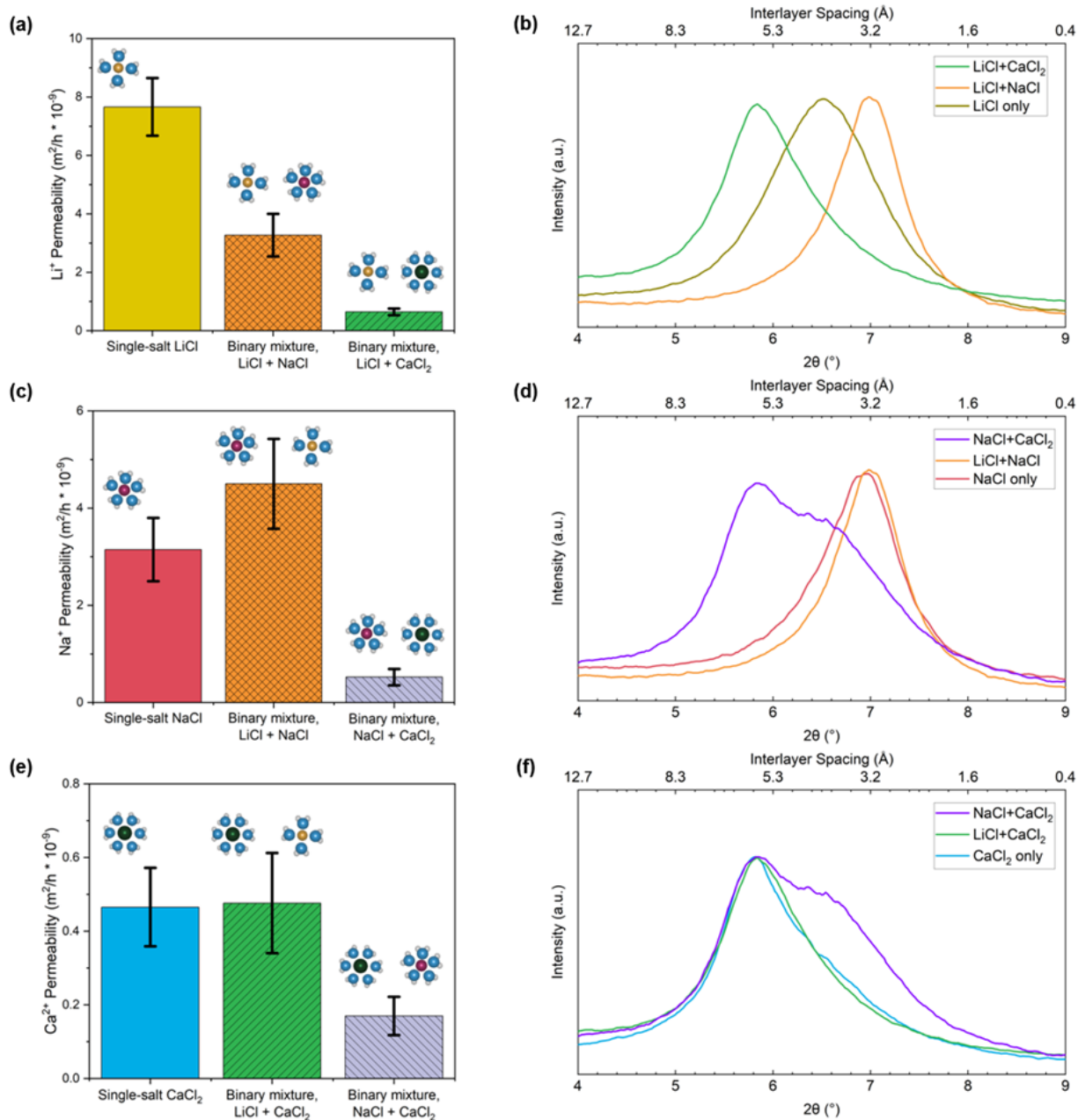


Figure S3: Side-by-side comparison of ion permeability with XRD peak shifts in different ionic environments. Ion permeability changes are correlated with shifts in the (002) peak position for Li^+ (a–b), Na^+ (c–d), and Ca^{2+} (e–f) mixtures.

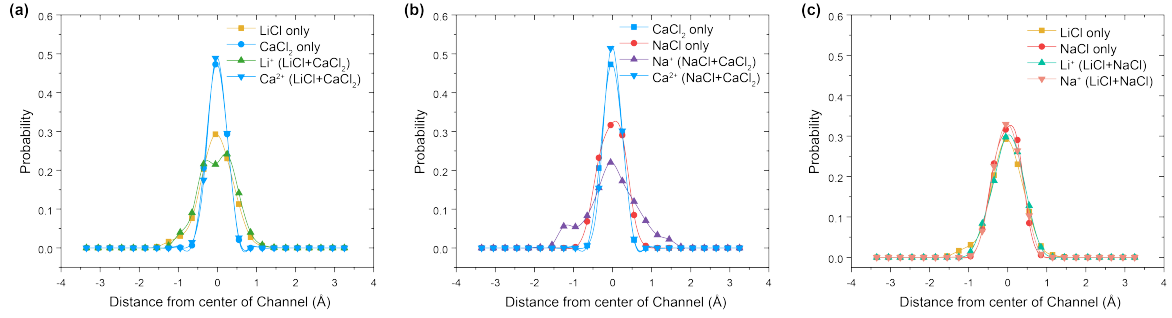


Figure S4: Simulated probability distribution of ions along the distance z from the center of the MXene channel under dry conditions: (a) Li^+ and Ca^{2+} , (b) Na^+ and Ca^{2+} , and (c) Li^+ and Na^+ , with interlayer spacing obtained from experimental measurements. Results for both single-salt solutions and binary mixtures are presented.

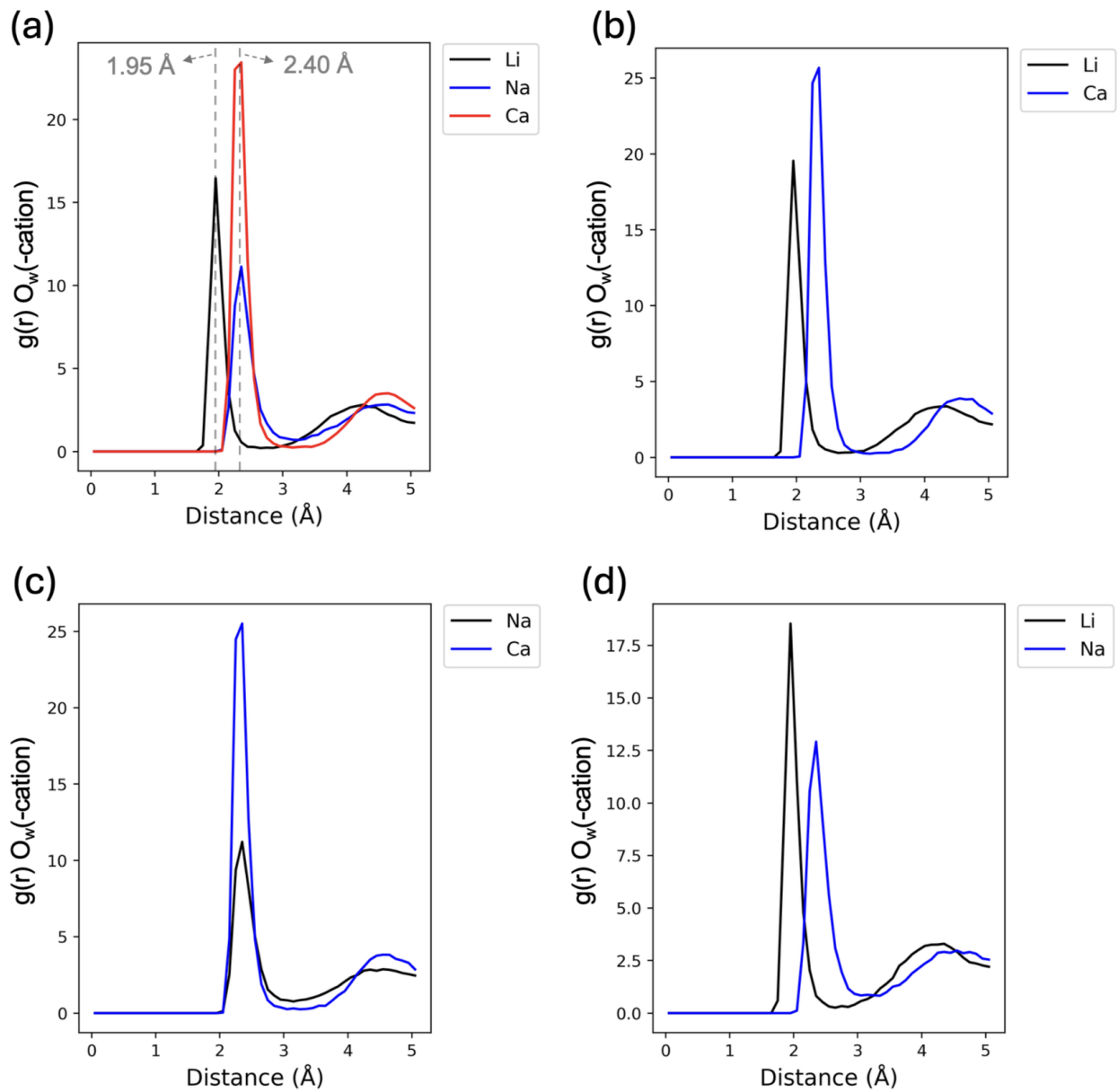


Figure S5: Radial distribution functions, $g(r)$, of water oxygen atoms (O_w) around cations from MD simulations. Results are shown for (a) single-salt solutions and (b–d) binary mixtures: (b) Li^+/Ca^{2+} , (c) Na^+/Ca^{2+} , and (d) Li^+/Na^+ , under wet conditions.

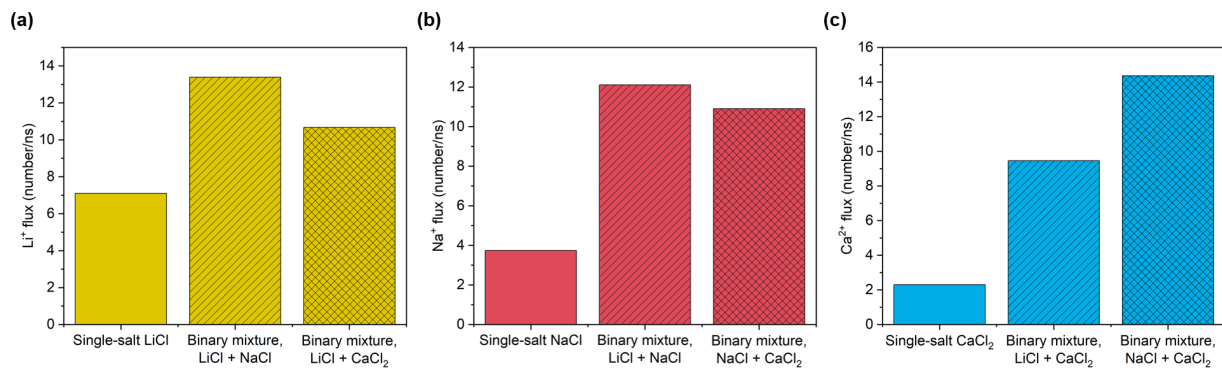


Figure S6: MD-simulated ion flux for (a) Li^+ , (b) Na^+ , and (c) Ca^{2+} ions, evaluated using wet interlayer spacing.

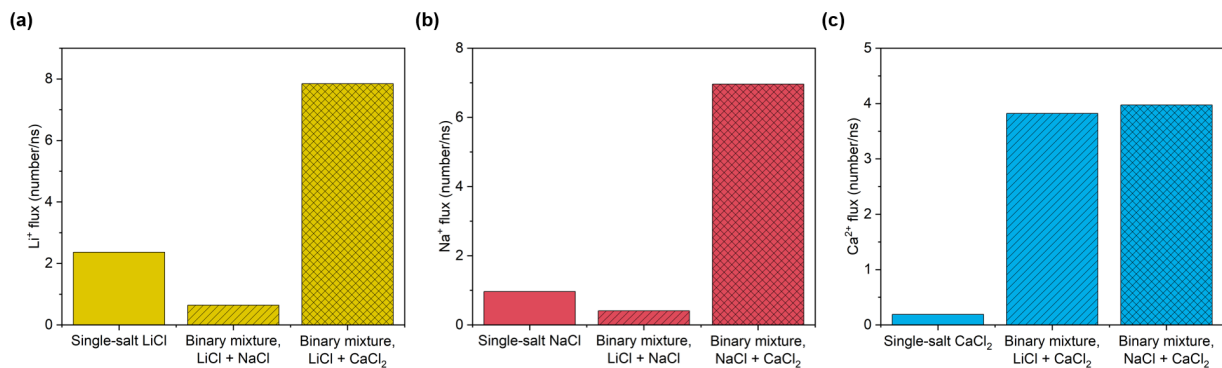


Figure S7: MD-simulated ion flux for (a) Li^+ , (b) Na^+ , and (c) Ca^{2+} ions, evaluated using dry interlayer spacing.

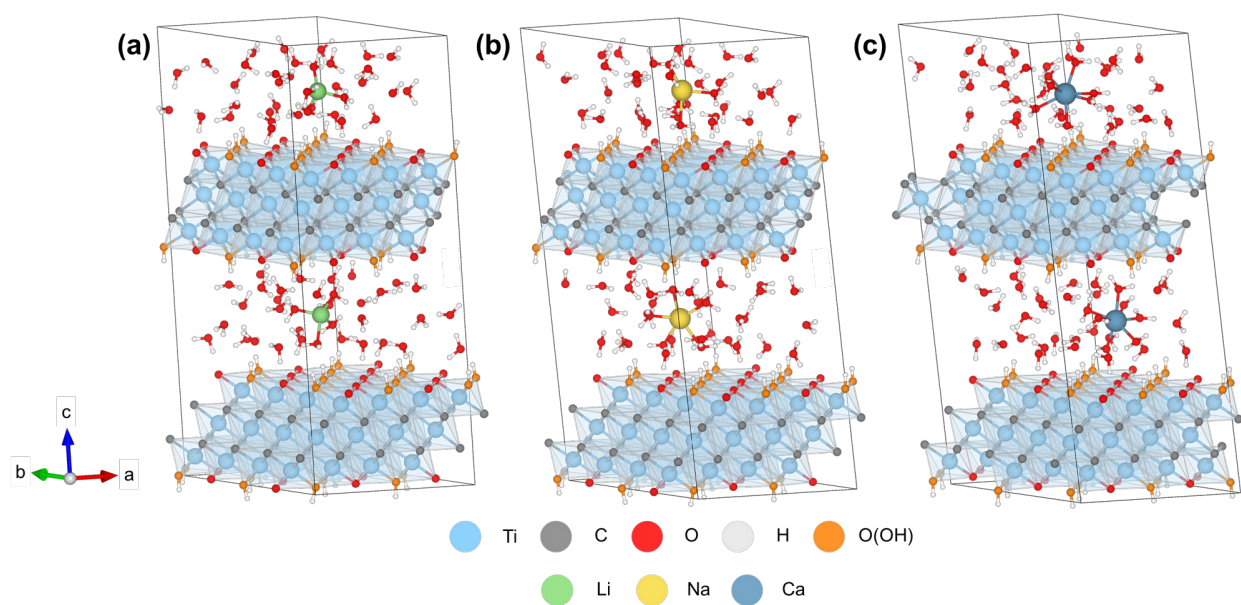


Figure S8: Optimized structures of $\text{Li}_{0.06}\text{Ti}_3\text{C}_2\text{OOH}\cdot 2.19\text{H}_2\text{O}$ (a), $\text{Na}_{0.06}\text{Ti}_3\text{C}_2\text{OOH}\cdot 2.19\text{H}_2\text{O}$ (b), and $\text{Ca}_{0.06}\text{Ti}_3\text{C}_2\text{OOH}\cdot 2.19\text{H}_2\text{O}$ (c).

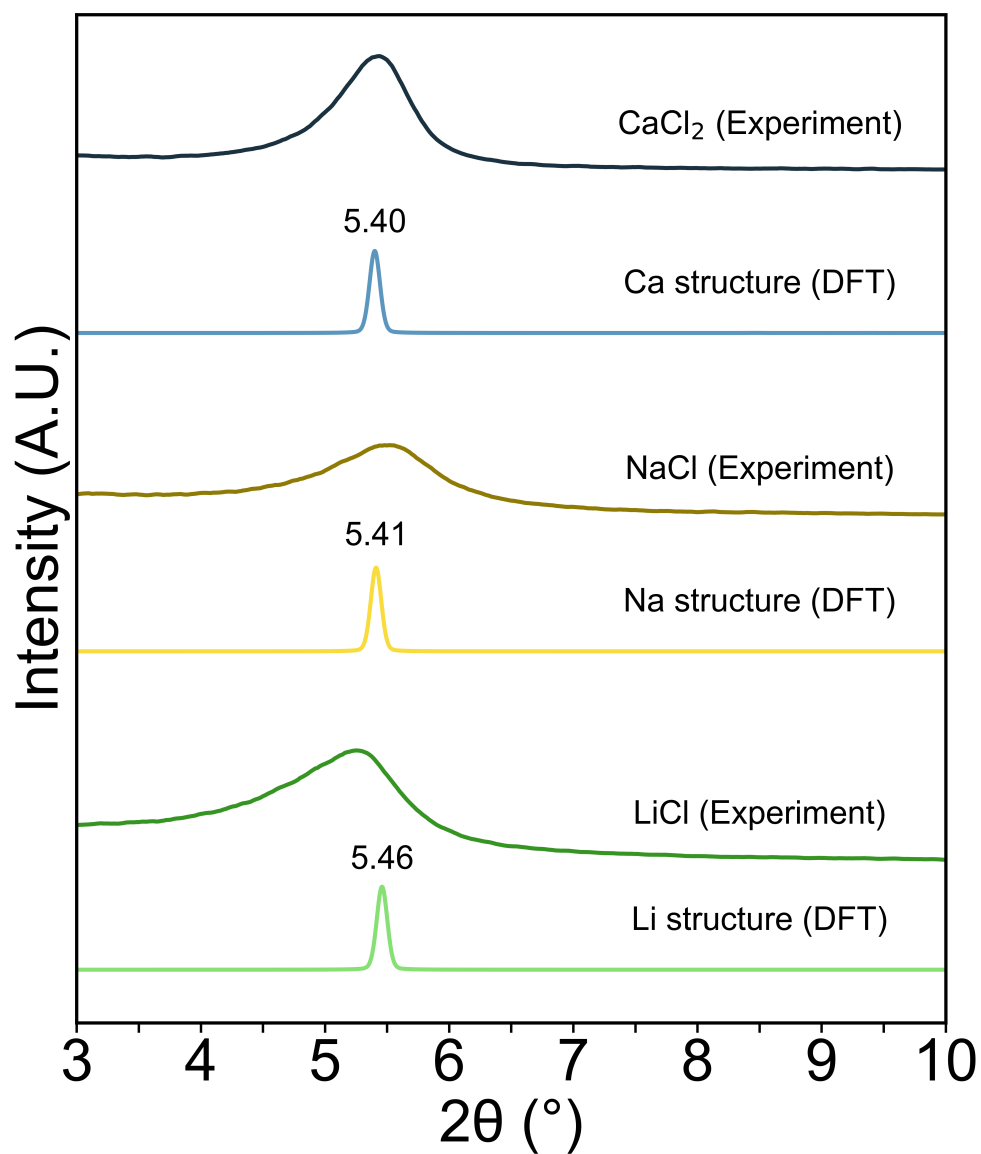


Figure S9: Comparison of XRD peaks between experimental and DFT calculated crystal structures. Pseudo-Voigt was applied to both experimental and DFT spectra.

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